

Ostrom 1990 Claude Guide

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1 Central Claims

- The “**tragedy of the commons**” (Hardin 1968) is not an inevitable fate of shared resources but a description of what happens to an **open-access resource**—one with *no* institutions governing use. Ostrom’s central move is to distinguish **open access** (no defined group of users, no rules) from **common property regimes** (a defined group of appropriators governed by rules they often devise themselves). Commons *can* be governed; the tragedy is a failure of institutions, not a property of the resource itself.
- The two policy prescriptions that dominate economics and policy analysis for common-pool resource (CPR) problems—**centralized government control** (the “Leviathan” solution, following Hobbes) and **privatization** (dividing the resource into individually owned parcels)—are neither necessary nor sufficient to avert overuse. Both can work in some settings and fail badly in others; neither is uniquely optimal, and both are frequently proposed without empirical support.
- A third path exists and is empirically common: **self-organized, self-governed institutions** created and enforced by the appropriators themselves, operating without external Leviathan enforcement or full privatization. Many such systems have persisted for centuries, successfully managing forests, irrigation systems, fisheries, and grazing lands.
- Rational, self-interested individuals *can* solve collective action problems in CPR settings when they can (1) communicate, (2) devise their own rules tailored to local conditions, (3) monitor each other’s compliance, and (4) enforce rules through graduated sanctions—contra the standard prediction of the collective action/free-rider models (Olson 1965; Hardin 1968) that voluntary cooperation without external enforcement will unravel.
- Successful long-enduring CPR institutions share a common set of **eight design principles** (detailed below). These are not a bureaucratic blueprint imposed from outside but regularities that Ostrom induces from comparative case analysis of institutions that have survived, versus those that have failed.
- Institutions matter, but they are not costless: appropriators face **second-order collective action problems**—the costs of designing, monitoring, and enforcing rules, and of monitoring the monitors themselves. Solving the first-order CPR dilemma (overuse) requires solving these second-order problems of institutional supply and credible commitment.
- The broader theoretical payoff is a rejection of monocausal, one-size-fits-all institutional prescriptions (whether market-based or state-based) in favor of a framework attentive to the specific attributes of resources, communities, and the rules-in-use that link them—an early and influential statement of what becomes **polycentric governance** theory.

2 Core Case Studies and Empirical Strategy

- **Method:** Ostrom uses structured, focused comparison across a small set of long-enduring, self-governed CPR institutions and a smaller set of failure cases, supplemented by theoretical modeling (game theory, especially iterated prisoner's dilemma and assurance games) and a broader survey of the CPR literature (irrigation, fisheries, forestry, groundwater). This is a most-different-systems design aimed at inducing common institutional features across otherwise very different resources, cultures, and eras.
- **Swiss alpine grazing commons (Törbel and other villages):** Communal ownership and management of high-altitude grazing meadows for over 500 years, coexisting with private ownership of arable land and houses in the same villages—demonstrating that commons and private property are not mutually exclusive and that villagers deliberately chose communal tenure for the grazing resource because it fit the resource's characteristics (high variance, difficulty of subdividing/monitoring boundaries).
- **Japanese communal land and forests (iraiichi):** Village-managed commons for grasses, fodder, fertilizer, and firewood, governed by detailed local rules on entry, harvesting seasons, and quotas, enduring for centuries under evolving but locally rooted institutions.
- **Spanish *huertas* (Valencia, Murcia, Alicante):** Centuries-old (some dating to Moorish administration) farmer-governed irrigation associations with elected officials (e.g., the Tribunal de las Aguas in Valencia), graduated fines for water theft, and locally devised rules for allocating water during scarcity—among the longest continuously operating self-governed CPR institutions Ostrom documents.
- **Philippine *zanjera* irrigation communities (Ilocos Norte):** Farmer-built and farmer-managed irrigation systems in which labor contributed to constructing/maintaining canals is directly linked to water rights (shares), aligning the provision problem (who builds/maintains) with the appropriation problem (who gets water) through the internal rule structure itself.
- **Failure/contrast cases:**
 - **California groundwater basins** (e.g., West Basin, Raymond Basin) prior to institutional reform: unregulated pumping led to overdraft and saltwater intrusion; appropriators eventually organized (with courts and special districts) to create governance institutions—illustrating both the tragedy under open access *and* the capacity of appropriators to self-organize once stakes were clear, albeit with significant transaction costs and reliance on external legal/judicial institutions.
 - **Turkish inshore (Alanya) fisheries—contrasted as an initial failure case that Ostrom also uses (in related work) as a success once local fishers devised a rotating-spot allocation system;** the broader point is that fisheries without such locally devised rules (as in several other failed fishery cases Ostrom surveys) suffer from severe overcapitalization and stock depletion.

- Additional failure examples (Sri Lankan fisheries, some groundwater basins, some grazing commons that collapsed under population pressure or external shocks) illustrate that self-governance is *not automatic*—it depends on the presence of the design principles, and its absence (or breakdown, e.g., due to externally imposed rule changes, rapid demographic/market change, or heterogeneity swamping trust) predicts failure.
- **The comparative logic:** By pairing enduring successes with failures involving similar resources, Ostrom isolates institutional features (not resource type, not culture, not wealth) as the operative variable distinguishing sustainable CPR governance from tragedy.

3 Core Works Cited

- **Garrett Hardin**, “The Tragedy of the Commons” (1968, *Science*)—the foil for the entire book. Hardin’s herders grazing a shared pasture to ruin is Ostrom’s point of departure; she argues Hardin actually modeled an open-access resource, not a commons with institutions, and that his own suggested solutions (coercive government or private property) are overgeneralized.
- **Mancur Olson**, *The Logic of Collective Action* (1965)—the foundational statement of the free-rider problem that CPR governance must overcome; Ostrom’s entire project can be read as identifying the empirical and theoretical conditions under which Olson’s pessimistic prediction fails to hold.
- **Thomas Hobbes**, *Leviathan* (1651)—the intellectual source of the “external sovereign enforcement” solution that Ostrom labels the “Leviathan” prescription and subjects to critique.
- **H. Scott Gordon** (1954) and **Anthony Scott** (1955)—foundational bioeconomic fisheries models formalizing the overexploitation logic later popularized by Hardin; part of the theoretical lineage Ostrom engages on CPR overuse.
- **James Buchanan and Gordon Tullock**, *The Calculus of Consent* (1962), and the broader **Bloomington School/public choice** tradition (Vincent Ostrom)—the methodological and theoretical home base for Ostrom’s approach: rational-choice institutionalism applied to self-governance and polycentricity.
- **Vincent Ostrom, Charles Tiebout, and Robert Warren**, “The Organization of Government in Metropolitan Areas” (1961)—origin of the concept of **polycentricity** (multiple, overlapping centers of decision-making authority) that underlies Ostrom’s institutional pluralism.
- **Robert Axelrod**, *The Evolution of Cooperation* (1984)—game-theoretic grounding for how repeated interaction, reciprocity, and reputational monitoring sustain cooperation

without central enforcement; directly informs Ostrom’s account of monitoring and graduated sanctions.

- **Robert Netting**, *Balancing on an Alp* (1981)—ethnographic study of the Swiss alpine commons (Törbel) that supplies much of Ostrom’s Swiss case material and her argument that commons and private property coexist by design, matched to resource attributes.
- **Prisoner’s Dilemma / Assurance Game literature**—Ostrom uses noncooperative game theory not to predict inevitable defection but to model how institutional rules change the payoff structure facing appropriators, converting a prisoner’s dilemma into an assurance game once monitoring and sanctioning are credible.

4 Chapter 1: Reframing the Problem—Beyond Markets and States

- **Three influential models predict CPR overuse:** (1) Hardin’s tragedy of the commons; (2) the prisoner’s dilemma game; (3) the logic of collective action (Olson). All three assume rational, self-interested individuals lack the capacity to change the rules governing their own interaction—Ostrom calls this the assumption that appropriators are trapped, unable to communicate or contract.
- **Ostrom’s reframing:** The question is not “markets or states?” but rather how institutional arrangements—formal and informal rules-in-use—shape the incentives facing appropriators of a shared resource. A resource’s physical characteristics alone do not determine the outcome; the governing institution does.
- **Definitional groundwork: Common-pool resources (CPRs)** are defined by two attributes: (1) **high subtractability/rivalness** (one user’s extraction subtracts from what is available to others, e.g., fish caught, water diverted) and (2) **difficulty/costliness of excluding** potential beneficiaries. This combination distinguishes CPRs from public goods (low subtractability, difficult exclusion) and toll/club goods (low subtractability, easy exclusion) and private goods (high subtractability, easy exclusion).
- **Two distinct collective action problems:** **appropriation** problems (how to allocate a limited flow of resource units, e.g., fish, water, fodder, to avoid overuse/rivalrous depletion) and **provision** problems (how to provide, maintain, or improve the resource system/infrastructure itself, e.g., canal maintenance, restocking), which involve underinvestment absent enforceable rules. Institutions must solve both.

5 Chapter 2: An Institutional Approach to the Commons

- **Institutions as rules-in-use:** Ostrom defines institutions as the sets of working rules used to determine who is eligible to make decisions, what actions are permitted or constrained, what procedures must be followed, what information must be provided, and what payoffs will be assigned to individuals depending on their actions—distinct from formal law on the books.
- **Rational-choice institutionalism:** Ostrom retains methodological individualism and rational, self-interested actors (contra romantic communitarian readings of “the commons”) but shows that rule structures—not altruism—explain successful cooperation. This is a critical distinguishing feature from naïve communitarian celebrations of “the commons.”
- **Critique of panaceas:** Both the privatization and centralization prescriptions treat institutional design as a technical, one-shot, top-down problem solvable by an omniscient external designer. Ostrom rejects this as unrealistic: information is costly and dispersed, monitoring and enforcement are themselves costly collective goods, and the same formal rule can produce different real-world outcomes depending on local physical, cultural, and economic context.
- **Three worlds framework:** Ostrom systematically works through why (a) the “Leviathan”/centralized-government solution, (b) unilateral private property division, and (c) purely voluntary cooperation without any rule change (Olson’s baseline) all face specific empirical and theoretical limits—setting up the case for self-organized alternatives explored in Chapters 3–5.

6 Chapter 3: Analyzing Long-Enduring, Self-Organized CPR Institutions

- This chapter presents the detailed case studies (Swiss/Japanese commons; Spanish *huer-tas*; Philippine *zanjeras*) and inductively derives the institutional regularities common to all of them despite vast differences in culture, resource type, and era.
- **The famous eight design principles** for robust, long-enduring CPR institutions:
 1. **Clearly defined boundaries:** Both the boundaries of the resource itself and the individuals/households with rights to appropriate from it are clearly specified, excluding outsiders from unauthorized use.
 2. **Congruence between appropriation/provision rules and local conditions:** Rules restricting time, place, technology, and/or quantity of resource units are tailored to local ecological and social conditions, and the distribution of benefits is

roughly proportional to the costs (labor, materials, money) appropriators are required to contribute.

3. **Collective-choice arrangements:** Most individuals affected by operational rules can participate in modifying those rules—rules are not imposed unilaterally from outside or by a narrow elite, which increases the likelihood rules fit local circumstances and are perceived as fair.
 4. **Monitoring:** Monitors, who actively audit CPR conditions and appropriator behavior, are accountable to the appropriators themselves or are appropriators themselves—solving the “who guards the guardians” second-order problem by embedding monitoring incentives in the rule structure (e.g., monitors are rotated appropriators, or paid from resource proceeds and answerable to the group).
 5. **Graduated sanctions:** Appropriators who violate rules receive graduated sanctions (depending on the seriousness and context of the offense) from other appropriators, officials accountable to them, or both—rather than either zero enforcement or immediate severe punishment. Graduation preserves proportionality and allows for honest mistakes while still deterring repeated violation.
 6. **Conflict-resolution mechanisms:** Appropriators and officials have rapid access to low-cost local arenas to resolve conflicts among themselves or with officials.
 7. **Minimal recognition of rights to organize:** The rights of appropriators to devise their own institutions are not challenged by external governmental authorities—external authorities (the state) tolerate or explicitly recognize local self-governance rather than overriding it.
 8. **Nested enterprises** (for CPRs that are part of larger systems): Appropriation, provision, monitoring, enforcement, conflict resolution, and governance activities are organized in multiple layers of nested enterprises, with local institutions embedded within (and coordinated with) larger-scale institutions—an application of the polycentricity concept to CPR governance at scale.
- **Why these principles work together:** They jointly solve the problem of **credible commitment**—appropriators will only restrain their own use if they believe others will too, and they will only believe others will if they can *observe* compliance (monitoring) cheaply, *trust* that violations are punished proportionately and fairly (graduated sanctions, conflict resolution), and *participate* in shaping rules they perceive as legitimate and well-matched to their situation (congruence, collective choice). Absent any one principle, the whole structure is vulnerable to erosion.

7 Chapters 4–5: Institutional Change, Failure, and Fragility

- **Institutional change is not automatic or costless:** Ostrom analyzes cases (e.g., California groundwater basins) where appropriators had to invest heavily—in scientific study,

litigation, negotiation, and new organizational forms (mutual water companies, special water districts)—before arriving at workable institutions. Successful institutional supply is itself a collective action problem requiring leadership, low initial transaction/information costs, and, often, external legal or governmental support to enforce agreements once reached.

- **Why institutions fail:** Common causes of breakdown include rapid population growth or migration outpacing rule adaptation, sudden increases in resource value (attracting outside claimants and undermining boundary rules), heterogeneity of appropriators eroding trust and common understanding, and external authorities imposing rule changes that ignore or override local design-principle-consistent arrangements (e.g., nationalizing forests or fisheries and displacing functioning local rules with poorly enforced central regulations).
- **The role of the state is not simply antagonistic:** Ostrom’s argument is not anti-state; central governments can support self-governance (principle 7, external recognition) or can destroy it (by asserting exclusive control without the monitoring/enforcement capacity to actually manage the resource, producing a worse outcome than either full centralization or full local control alone—a point later echoed in critiques of “fortress conservation” and top-down natural-resource nationalization in the developing world).
- **Scale and nesting:** Larger, more complex CPRs (river basins, regional fisheries, trans-boundary forests) require nested institutional arrangements—local rule-making embedded within regional and national coordinating institutions—because no single scale of organization can solve every relevant collective action problem (local appropriation rules vs. basin-wide allocation vs. national environmental regulation).

8 Chapter 6: A Framework for Analysis of Self-Organizing and Self-Governing CPRs

- **The Institutional Analysis and Development (IAD) framework:** Ostrom formalizes her approach into a general analytic framework distinguishing the **action arena** (participants and an action situation), the **rules-in-use** that structure it, and **physical/material conditions** and **community attributes** as exogenous variables shaping the arena—becoming one of the most widely used frameworks in institutional analysis across political science, economics, and environmental studies.
- **Second-order collective action problems, formalized:** Ostrom explicitly identifies the challenge of explaining why appropriators would bear the costs of *creating* monitoring and enforcement institutions (a public good in itself) in the first place—a problem logically prior to, and just as serious as, the first-order overuse problem. Her answer emphasizes gradual institutional bootstrapping: small initial agreements with low enforcement costs build trust and reputational capital that make larger institutional investments feasible.
- **Conditions favoring successful self-organization:** Include (but are not limited to)

the resource and appropriators being spatially bounded and observable; reliable and valid indicators of resource conditions being available at reasonable cost; resource flow being relatively predictable; appropriators depending substantially on the resource for their livelihoods (raising the stakes of investing in institutions); appropriators sharing a common understanding of the resource and its dynamics; low discount rates (concern for the future); and low-cost, effective conflict-resolution mechanisms available locally.

9 Methodological Contributions and Limitations

- **Contribution: bridging rational choice and institutional/comparative analysis:** Ostrom shows that rigorous rational-choice modeling and rich comparative case analysis are complements, not substitutes—formal game-theoretic logic explains *why* certain rule configurations solve credible-commitment problems, while comparative cases identify *which* configurations actually recur among survivors.
- **Contribution: an inductive, bottom-up theory of institutions:** Rather than deriving institutional prescriptions from abstract efficiency criteria, Ostrom derives the design principles empirically from what has actually endured—an important methodological corrective to purely deductive policy analysis (and to purely normative celebrations of “community”).
- **Contribution: the IAD framework and IASD/polycentricity research program:** Both spawned enormous downstream research programs (the Ostrom Workshop at Indiana University, the International Forestry Resources and Institutions [IFRI] program) applying and testing the design principles across thousands of cases worldwide.
- **Limitation: selection on the dependent variable:** The core cases are chosen because they are long-enduring successes (and a smaller number of clear failures); critics note the design principles are therefore somewhat tautological (institutions that endure tend to have features that help them endure) and that the causal weight of each principle, and whether all eight are jointly necessary, is not established with the same rigor across cases.
- **Limitation: scope conditions:** The clearest successes are small-to-medium-scale, relatively homogeneous communities with long time horizons and stable resource/technology conditions. Ostrom is candid that the framework is less obviously suited, without modification, to large-scale, highly heterogeneous, or rapidly changing resource systems (e.g., global commons like climate change, ocean fisheries beyond coastal zones)—a limitation she and collaborators addressed in later work on polycentric governance of global commons.
- **Limitation: power and distribution:** The book pays comparatively little attention to internal inequality, gender, and power asymmetries within appropriator communities—critics (in subsequent commons scholarship) note that “community” rule-making can encode and reproduce unequal access (e.g., elite capture of collective-choice arrangements) even while satisfying the formal design principles.

- **Limitation: exogenous shocks and the state:** The framework is better at explaining institutional persistence under relatively stable conditions than at predicting how institutions will (or won't) adapt to rapid exogenous shocks—climate change, market integration, state consolidation—that can outpace local rule adaptation.

10 Connections to the Broader Literature

- **Olson 1965:** The book's central theoretical target. Olson's *Logic of Collective Action* predicts that large, latent groups of rational self-interested individuals will fail to provide themselves with collective goods absent coercion or selective incentives. Ostrom does not refute Olson's logic so much as specify empirically grounded conditions under which small-to-medium, spatially bounded, resource-dependent groups escape the Olsonian trap by supplying their own monitoring, sanctioning, and rule-making institutions—turning “the group” from Olson's abstraction into a concrete, historically and institutionally specific set of appropriators.
- **Hardin's Tragedy of the Commons and the “Leviathan”/privatization consensus:** Ostrom's most direct polemical target. She reframes Hardin's parable as a model of *open access*, not commons management, and empirically undermines the assumed superiority of the two standard policy responses (centralize or privatize) that Hardin's article helped canonize in economics and policy circles.
- **Huntington 1968:** Both authors are centrally concerned with the gap between social/economic change and institutional capacity, and both reject purely formal or purely economic accounts of order; where Huntington focuses on national political institutionalization amid mobilization, Ostrom focuses on local resource-governance institutionalization amid collective action pressure. Both treat institutions as artifacts that must be actively built and maintained, not automatic byproducts of development.
- **Dahl 1972:** Dahl's polyarchy requires organized pluralism and the toleration of contestation; Ostrom's polycentricity (multiple, overlapping, locally accountable centers of governance) is a resource-governance analogue to Dahl's institutional pluralism, and both authors resist single-center (Leviathan-style) solutions to problems of collective governance.
- **Moore 1966:** Moore's structural-historical analysis of how different class coalitions and agrarian structures produced divergent paths to modernity shares Ostrom's skepticism of universal, one-size-fits-all developmental/institutional prescriptions, though Moore works at the level of national political trajectories/revolutions while Ostrom works at the level of micro-institutional design; both insist that context (agrarian class structure for Moore; resource and community attributes for Ostrom) conditions which macro-level solutions are viable.
- **Putnam 1993:** Putnam's account of social capital and civic community as the substrate that enables effective collective governance in Italian regions closely parallels Ostrom's

emphasis on trust, monitoring, and reciprocity as the glue that allows appropriators to sustain cooperation; Ostrom’s design principles can be read as the micro-institutional machinery through which Putnam’s civic norms and networks are operationalized in a specific policy domain (resource management) rather than general government performance. Both push back on purely economic/formal-institutional accounts.

- **Weber 1930 and rational-legal authority:** Ostrom’s appropriator-devised rule systems (elected water-masters, codified sanctions, recordkeeping) exhibit many features of Weberian rational-legal authority emerging endogenously at the local level, without reliance on the modern bureaucratic state Weber typically has in mind—an interesting complication of the Weberian association between rational-legal order and centralized bureaucracy.
- **Cox 1997:** Cox’s work on electoral systems and coordination among strategic actors shares Ostrom’s rational-choice-institutionalist method—both explain macro-level patterns (party systems for Cox; resource sustainability for Ostrom) as equilibrium outcomes of individually rational responses to specific institutional rules, rather than as products of culture or exogenous preferences alone.
- **Lijphart 1999:** Lijphart’s consensus vs. majoritarian democracy typology and Ostrom’s critique of centralized/unitary (Leviathan) versus fragmented/privatized solutions both argue that institutional design is not one-dimensional and that hybrid, power-sharing, or polycentric arrangements often outperform either extreme—a shared skepticism of single “best” institutional forms.
- **Huber and Shipan 2002:** Huber and Shipan’s analysis of how legislators write more or less discretion into statutes based on their ability to monitor and trust bureaucratic agents parallels Ostrom’s principles 4 and 5 (monitoring, graduated sanctions): both literatures treat the credibility of monitoring and enforcement as the central design variable that determines whether a principal (legislature; appropriator community) can safely delegate/cooperate.
- **Powell 2000:** Powell’s comparative account of how electoral rules shape the congruence between citizen preferences and policy outputs resonates with Ostrom’s design principle of **congruence** between rules and local conditions—both scholars argue institutional performance depends on fit between rule structure and the underlying environment, not on the intrinsic superiority of any single rule set.
- **Stokes 2013:** Stokes’s analysis of clientelism as a strategy of monitoring and sanctioning voters to enforce an exchange (votes for goods) is structurally analogous to Ostrom’s account of monitoring and graduated sanctions enforcing appropriator compliance—both show that sustaining a cooperative/exchange equilibrium among many dispersed individuals requires credible, low-cost monitoring and enforcement mechanisms, whether the “exchange” is votes-for-patronage or restraint-for-resource-access.
- **Svolik 2012:** Svolik’s analysis of authoritarian power-sharing as a problem of credible

commitment between dictators and elites (who monitors compliance, how are violations punished) mirrors Ostrom’s core problem of credible commitment among appropriators; both literatures treat institutions as devices for making cooperative promises self-enforcing in the absence of an external, impartial enforcer (or where the internal “enforcer” itself must be constrained).

- **Carey 2009:** Carey’s work on legislative voting and party discipline, like Ostrom’s account of collective-choice arrangements (principle 3), addresses how institutional rules determine who gets a voice in modifying the rules that bind a group, and how that participation shapes compliance and legitimacy.
- **Polycentric governance and later work:** Ostrom’s 1990 framework directly generated her later, Nobel-cited work (with Vincent Ostrom and collaborators) on polycentric governance of complex social-ecological systems, including climate change governance—arguing that multiple, overlapping, locally rooted institutions operating at different scales can outperform any single global (Leviathan) climate regime, extending the book’s central lesson from village commons to planetary commons.